Celestial Mechanics

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Contents

Prerequisite Knowledge

- Newton's laws of motion, and basic circular motion (centripetal force)
- Vectors: dot and cross products, and their geometric meaning
- Basic differentiation and integration (power rule, chain rule)
- Comfort with polar coordinates is helpful, but is built up from scratch here

1 Newton's Laws of Gravitation

Let's start with something almost embarrassingly simple, and then notice why it's actually one of the most astonishing statements ever made about nature.

Newton's claim was this: *every* pair of masses in the universe pulls on each other. Not just the Earth and the Moon, not just planets and the Sun — literally you and the chair you're sitting in are gravitationally attracted to each other right now. The force is tiny, of course, because gravity is an extraordinarily weak force compared to, say, the electric force between two charges. But it's there, and Newton wrote down exactly how strong it is:

$$\vec{F} = -\frac{GMm}{r^2}\,\hat{r}.$$

Read this slowly. M and m are the two masses. r is the distance between their centers. And G is a tiny number, $6.674 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}$, whose smallness is precisely why you don't feel yourself being pulled toward your chair — the masses involved (a person, a chair) are just too small for this feeble force to be noticeable. It takes something the size of a *planet* to make gravity impressive.

The minus sign and the $\hat{r}$ (a unit vector pointing from M toward m) are just bookkeeping to say: the force always points *back* toward the other mass. Gravity never pushes, it only pulls. This is worth sitting with for a second, because it's not obvious a priori that nature had to work this way — electric charges, after all, can either attract or repel depending on their sign. Mass has no such sign. There's no such thing as "negative mass" that would make gravity repulsive. Every mass, no exceptions, only attracts.

Now here's the part that made Newton's law so revolutionary at the time: the very same $1/r^2$ law that makes an apple fall from a tree also keeps the Moon in orbit around the Earth, and the Earth in orbit around the Sun. Before Newton, people genuinely believed that "the heavens" ran on completely different rules than "the Earth" did — that celestial motion was some perfect, eternal, circular clockwork, fundamentally unlike the messy business of things falling down here on the ground. Newton's insight was that there is no difference. The Moon is falling, all the time, exactly the way an apple falls — it's just moving sideways fast enough that as it falls, the ground (in this case, the curved surface implied by the Earth's gravity) keeps "falling away" underneath it at the same rate. That's what an orbit *is*: a continuous fall that never quite lands.

Gravitational Fields

It's often convenient to stop thinking about the force on a *specific* mass m, and instead ask: what would the force be on *any* mass placed at this point in space? Dividing the force by the test mass m gives us the gravitational field,

$$\vec{g}(\vec{r}) = \frac{\vec{F}}{m} = -\frac{GM}{r^2}\hat{r},$$

which depends only on the source M and the location, not on whatever test mass we imagine placing there. Think of it like a landscape of "pull" that exists in space whether or not anything is there to feel it — much like how a river has a current everywhere, whether or not there's a boat on it to be swept along.

One beautiful and enormously useful fact about gravitational fields: they add up. If you have several masses scattered around, the total field at any point is just the vector sum of what each mass would contribute on its own, as if the others weren't even there. This is called the **principle**

of superposition, and it's what lets us compute the field of complicated shapes — spheres, rings, disks, even lumpy asteroids — by imagining them chopped into a huge number of tiny point masses and adding up all their individual pulls.

A Word on "Central" Forces

Notice something about gravity's structure: its strength depends *only* on the distance r, and its direction is *always* along the line connecting the two masses. A force like this — depends only on distance, always points along the radius — is called a **central force**. It sounds like a small, technical observation. It is anything but. As we're about to see, this one property, all by itself, is responsible for essentially everything Kepler discovered about planetary motion, centuries before anyone knew *why* his laws were true. Kepler found the rules by painstakingly staring at Tycho Brahe's observational data; we're going to derive the same rules by staring at one simple structural fact about the force itself.

2 Kepler's Laws for Circular and Non-Circular Orbits

2.1 Why Being "Central" is Such a Big Deal

Here's a question worth pausing on: why does a planet's orbit stay in a flat plane at all? Why doesn't the Earth, say, gradually wander up and down out of the plane it orbits in, tracing some complicated three-dimensional squiggle around the Sun?

The answer comes from a quantity called **angular momentum**, defined as

$$\vec{L} = m(\vec{r} \times \vec{v}).$$

If you haven't spent time with cross products, here's the idea: $\vec{r} \times \vec{v}$ produces a new vector that is perpendicular to *both* $\vec{r}$ and $\vec{v}$ — think of it like the axle of a spinning wheel, pointing straight out of the plane that the wheel spins in, at a right angle to everything happening in that plane.

Now let's ask how $\vec{L}$ changes with time:

$$\frac{d\vec{L}}{dt} = m\left(\frac{d\vec{r}}{dt} \times \vec{v}\right) + m\left(\vec{r} \times \frac{d\vec{v}}{dt}\right) = m(\vec{v} \times \vec{v}) + \vec{r} \times (m\vec{a}) = \vec{r} \times \vec{F}.$$

The first term vanished because any vector crossed with itself is always zero (there's no "twist" between a vector and its own direction). We're left with $\vec{r} \times \vec{F}$. And here's where centrality earns its keep: since the gravitational force $\vec{F} = F(r)\hat{r}$ points *along* $\vec{r}$, the two vectors are parallel, and the cross product of parallel vectors is zero. So:

$$\boxed{\frac{d\vec{L}}{dt} = 0.}$$

Angular momentum doesn't change. Not its size, not its direction. Ever. For as long as the force stays central.

Why does this force planarity? Because $\vec{L}$ is always perpendicular to $\vec{r}$ (that's built into the definition of the cross product), and $\vec{L}$'s *direction* never changes. So $\vec{r}$ is forever confined to lie in the one fixed plane that's perpendicular to this unchanging $\vec{L}$. If the planet ever tried to drift out of that plane, $\vec{r} \times \vec{v}$ would have to tilt, which would mean $\vec{L}$ changed direction — but we just showed it can't. The planet is, in a sense, trapped in its orbital plane by a conservation law, the same way a spinning ice skater who pulls her arms in doesn't just *decide* to spin faster — she's forced to, by the very same kind of conservation.

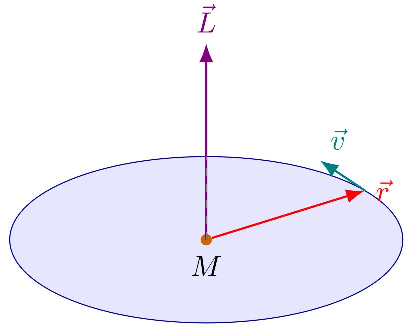

Figure 1: The orbital plane, pinned in place by the fixed direction of $\vec{L}$. The planet's whole trajectory — however complicated its shape — lives entirely in this one flat sheet.

2.2 Setting Up the Motion: Polar Coordinates

Since we now know the motion is confined to a plane, we only need two numbers to describe where the planet is: a distance r from the Sun, and an angle θ around it. This is a much more natural choice than x and y, because gravity itself only cares about r — so let's build our description of motion out of coordinates gravity actually respects.

The subtlety with polar coordinates is that the "unit vectors" themselves rotate as the planet moves — $\hat{r}$ always points from the Sun to wherever the planet currently is, so as the planet swings around, $\hat{r}$ swings with it. Working through the calculus of this rotation (a standard but slightly fiddly exercise) gives velocity and acceleration as:

$$\vec{v} = \dot{r}\hat{r} + r\dot{\theta}\hat{\theta}, \qquad \vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}.$$

Don't worry about memorizing this derivation — what matters is what happens next. Newton's second law, $\vec{F} = m\vec{a}$, for a purely radial force, has no tangential ($\hat{\theta}$) component at all. So the $\hat{\theta}$ part of $\vec{a}$ must be zero all by itself:

$$r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 \quad \Longleftrightarrow \quad \frac{1}{r}\frac{d}{dt}(r^2\dot{\theta}) = 0 \quad \Longrightarrow \quad r^2\dot{\theta} = \text{constant}.$$

Stop and notice something wonderful here: $mr^2\dot{\theta}$ is *exactly* the magnitude of the angular momentum L we met a moment ago. We didn't need to invoke angular momentum conservation as some separate fact handed down from above — it fell right out of writing Newton's second law in the coordinates that suit the problem. This is a recurring theme in physics: pick the right coordinates, and the conservation laws practically write themselves.

2.3 Kepler's Second Law: Equal Areas in Equal Times — for Any Shape of Orbit

Here's a lovely geometric fact hiding in that last equation. As the planet sweeps through a tiny angle $d\theta$ at radius r, it traces out a thin, almost-triangular sliver of area,

$$dA = \frac{1}{2}r^2\, d\theta$$

(picture a very thin pizza slice — its area is half the radius times the little arc it spans). Divide by dt:

$$\boxed{\frac{dA}{dt} = \frac{1}{2}r^2\dot{\theta} = \frac{L}{2m} = \text{constant.}}$$

This is Kepler's Second Law, and notice we've proven it for *any* orbit shape whatsoever — circular, elliptical, even the unbound parabolic and hyperbolic paths we'll meet shortly. It has nothing specifically to do with orbits being ellipses; it's a direct, unavoidable consequence of gravity being central, full stop.

The physical picture is worth internalizing: when a planet is close to the Sun (near perihelion), it's moving *fast*, sweeping through a wide angle very quickly. When it's far from the Sun (near aphelion), it moves *slowly*, crawling through only a small angle. But because the close approach compensates with a wide angle and small radius, while the distant part compensates with a narrow angle and large radius, the two effects trade off exactly, and equal areas get swept in equal times, always. This is precisely why comets — which have extremely eccentric orbits — spend the overwhelming majority of their orbital period crawling through the frigid outer solar system, and only a brief, blazing moment whipping around the Sun at perihelion.

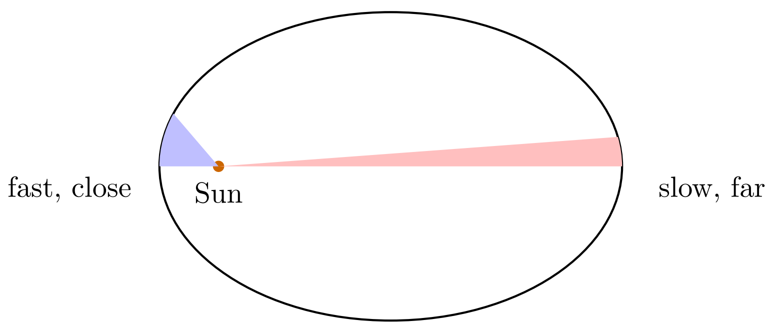

Figure 2: The two shaded wedges have equal area, and were swept out in equal time — one very compressed in angle but close to the Sun, the other spread wide in angle but very far away.

2.4 Circular Orbits First: Kepler's Third Law, the Easy Way

Before tackling the general, non-circular case, let's warm up with the simplest possible orbit: a perfect circle. Here, the planet's speed never changes, and the only job gravity has is to constantly redirect the planet's velocity so that it curves around instead of flying off in a straight line. That redirecting force is exactly what we call centripetal force, and gravity supplies it:

$$\frac{GMm}{r^2} = m\omega^2 r, \qquad \omega = \frac{2\pi}{T}.$$

Rearranging,

$$\boxed{T^2 = \frac{4\pi^2}{GM}r^3.}$$

Notice something curious already at this simple level: the bigger the orbit (r larger), the *longer* the period, but not in a simple linear way — it grows like $r^{3/2}$. Double the orbital radius, and the period doesn't just double; it grows by a factor of $2\sqrt{2} \approx 2.83$. This is why Mercury, tucked in close to the Sun, zips around in just 88 days, while Neptune, way out at the edge of the solar system, takes about 165 *years* to complete a single orbit.

2.5 Now the Real Question: What Shape is a Non-Circular Orbit?

Circles are easy, but nature rarely gives us anything so tidy. Real orbits are ellipses, sometimes quite eccentric ones. So the natural question is: given that gravity is a central, inverse-square force, what *shape* does the actual trajectory trace out? This is the derivation that let Newton finally explain, from first principles, something Kepler had only guessed at from data: that orbits are ellipses.

The trick is a clever change of variables. Instead of tracking r as a function of time t (which is messy, because both r and θ are changing simultaneously and in a coupled way), we're going to track $u \equiv 1/r$ as a function of the *angle* θ instead. Why would this help? Because we already know, from Kepler's second law, exactly how θ relates to time: $\dot{\theta} = L/(mr^2) = Lu^2/m$. This lets us convert any time-derivative into a θ-derivative.

Working this through (differentiating $r = 1/u$ with respect to t, using the chain rule and the relation above, twice), the messy radial equation of motion

$$m(\ddot{r} - r\dot{\theta}^2) = -\frac{GMm}{r^2}$$

transforms, after some genuinely satisfying algebra, into

$$\boxed{\frac{d^2u}{d\theta^2} + u = \frac{GMm^2}{L^2}.}$$

Take a moment to appreciate how remarkable this equation is. If you've ever studied a mass on a spring, or a pendulum swinging through small angles, you'll recognize this *exact* mathematical form — it's the equation for simple harmonic motion, just with θ playing the role that time usually plays! The messy, seemingly unrelated problem of planetary orbits has been quietly transformed into a problem we already know how to solve, one that a student meets in their very first weeks of studying oscillations.

The general solution to this "SHM in disguise" equation is a constant particular solution plus an oscillating piece:

$$u(\theta) = \frac{GMm^2}{L^2}\left[1 + e\cos(\theta - \theta_0)\right].$$

Choosing our angular reference point so that $\theta_0 = 0$ (measuring θ from the point of closest approach) and flipping back to $r = 1/u$:

$$\boxed{r(\theta) = \frac{p}{1 + e\cos\theta}}, \qquad p \equiv \frac{1}{k} = \frac{L^2}{GMm^2}.$$

This is, precisely, the equation of a conic section in polar form, with one focus at the origin. Depending on the value of e (which is set by the orbit's energy and angular momentum, not chosen freely), we get:

- $e = 0$: a perfect circle.
- $0 < e < 1$: an **ellipse**. This is Kepler's First Law, now not just observed but *derived*: bound orbits under gravity must be ellipses (or the special case of a circle).
- $e = 1$: a parabola — an orbit that is only just barely unbound, arriving at infinity with exactly zero leftover speed.

- $e > 1$: a hyperbola — a genuinely unbound trajectory, like a comet passing through once and never returning, or a spacecraft on a flyby.

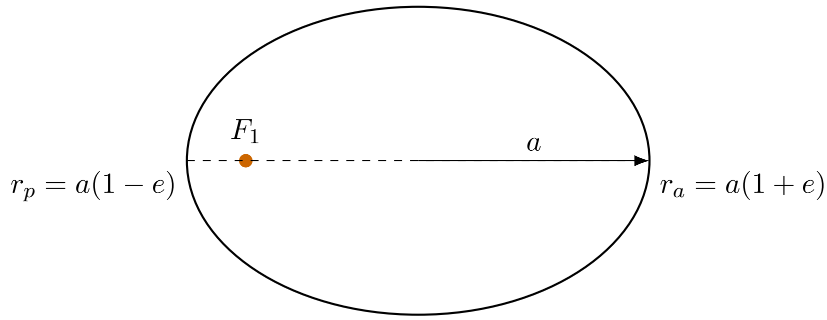

Figure 3: The Sun sits at a focus of the ellipse F_1 — not at the geometric center, a detail that surprises many people the first time they think carefully about it.

2.6 A Different Lens: Thinking in Terms of Energy

There's a second, equally illuminating way to understand the shape of an orbit, and it's worth learning both because different problems will suit different approaches. Instead of tracking the orbit's geometric shape directly, let's track its energy.

The gravitational potential energy comes from asking: how much work does it take to move a mass from radius r out to infinity, fighting against gravity the whole way? Integrating the force:

$$U(\infty) - U(r) = -\int_r^\infty \left(-\frac{GMm}{r'^2}\right) dr' = -\frac{GMm}{r}.$$

With the natural convention $U(\infty) = 0$ (two masses infinitely far apart don't interact, so there's no stored energy between them), we get the familiar

$$\boxed{U(r) = -\frac{GMm}{r}.}$$

The negative sign here is worth pausing on: it tells us that a bound system (planet orbiting Sun) has *less* total energy than two masses sitting infinitely far apart at rest. That makes intuitive sense — you'd have to *add* energy to the system to pull the planet away to infinity, so being bound must correspond to an energy deficit relative to being unbound.

Now, using $v^2 = \dot{r}^2 + r^2\dot{\theta}^2$ and the fact that $r^2\dot{\theta} = L/m$ is fixed, the total mechanical energy becomes

$$E = \frac{1}{2}m\dot{r}^2 + \underbrace{\frac{L^2}{2mr^2} - \frac{GMm}{r}}_{V_{\text{eff}}(r)}.$$

We've bundled the angular-momentum piece together with the potential energy into something called the **effective potential**, $V_{\text{eff}}(r)$. Why is this useful? Because now the problem *looks* exactly like a one-dimensional particle moving under some potential V_{eff} — as if the planet were a marble rolling around in a valley whose shape is given by this curve, with r playing the role of horizontal position. The genuinely two-dimensional orbit problem has been squashed down into a much more familiar-looking one-dimensional problem.

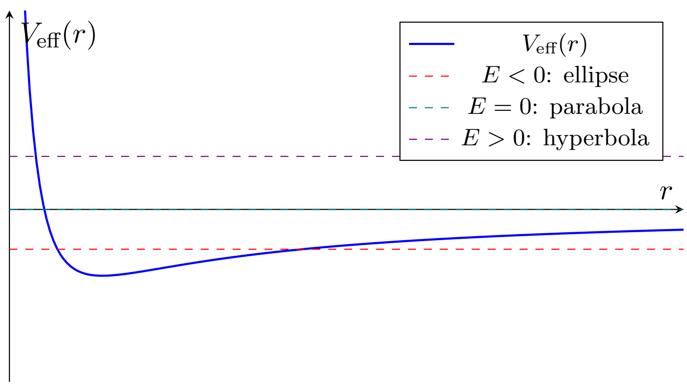

Figure 4: The effective-potential "valley." Wherever a horizontal energy line crosses the curve marks a turning point of the real orbit — the perihelion and aphelion distances. Notice the curve blows up to $+\infty$ as $r \to 0$: this is the angular-momentum barrier (sometimes called the centrifugal barrier), and it's the reason a planet with any angular momentum at all can never actually crash straight into the Sun — it always gets "turned around" first, no matter how close it gets.

Think about what this diagram is really telling you. If the total energy E is negative, the "marble" doesn't have enough energy to climb out of the valley on the right side (toward $r \to \infty$, where $V_{\text{eff}} \to 0$), so it's trapped, oscillating back and forth between two turning points forever — that's a bound, elliptical orbit. If E is exactly zero, the marble *just barely* makes it out to $r = \infty$ with no speed left over — a parabola. If E is positive, the marble sails out to infinity with speed to spare — a hyperbola. This single picture contains the entire classification of orbit types, and you didn't need to solve any differential equation to get it; you just needed to know the shape of one curve.

2.7 Back to Kepler's Third Law: Now for Any Ellipse

We derived Kepler's Third Law earlier for the special case of a circle. Let's now get the fully general result for an ellipse of any eccentricity, using the equal-areas result from earlier as our tool.

Over one full orbital period T, the radius vector must sweep out the *entire* area of the ellipse, which is πab (a standard geometric fact: an ellipse with semi-axes a and b has area πab, generalizing the familiar circle formula πr^2). Since the rate of area-sweeping is constant at $L/2m$,

$$\pi ab = \frac{L}{2m}T \quad \Longrightarrow \quad T = \frac{2m\pi ab}{L}.$$

Now we need to relate L back to the orbit's shape. Comparing $p = b^2/a$ (a standard geometric relation for ellipses) to our earlier expression $p = L^2/(GMm^2)$, we can solve for L in terms of the shape parameters, and substituting back in gives, after the dust settles,

$$\boxed{T^2 = \frac{4\pi^2}{GM}a^3.}$$

This is, remarkably, the *exact same formula* we found for a circular orbit, just with the circle's radius r replaced by the ellipse's semi-major axis a. The period doesn't care at all about the eccentricity — two orbits with wildly different shapes (one nearly circular, one a long thin sliver) but the same semi-major axis will complete their journeys in exactly the same amount of time. If that feels surprising, it should — it's one of the genuinely elegant surprises buried inside Newton's law.

2.8 The Vis-Viva Equation: Speed Anywhere on the Orbit

Here's a practical question a lot of orbital-mechanics problems boil down to: given that a spacecraft (or planet, or comet) is at some particular distance r from the central body, how fast is it moving *right now*?

We could, in principle, always go back to conserving energy and angular momentum separately every single time, but there's a shortcut worth having in your toolkit. It turns out (and can be checked directly by evaluating the energy at the perihelion point, where all the algebra simplifies nicely) that the total energy of *any* bound Keplerian orbit depends *only* on its semi-major axis, not on its eccentricity:

$$E = -\frac{GMm}{2a}.$$

Setting this equal to our general expression for energy, $E = \frac{1}{2}mv^2 - GMm/r$, and solving for v^2:

$$\boxed{v^2 = GM\left(\frac{2}{r} - \frac{1}{a}\right).}$$

This little equation is a real workhorse. Notice what it's telling you: speed depends on where you currently are (r) relative to the overall size of the orbit (a). Near perihelion, where r is smallest, the $2/r$ term dominates and speed is largest. Near aphelion, where r is largest, speed drops. This matches the intuition we already built from Kepler's Second Law — of course the planet moves fastest when closest and slowest when farthest, but now we have the precise quantitative relationship, not just the qualitative trend.

2.9 Escape Velocity, and What "Energy ≥ 0" Really Means

What if a planet, or a rocket, has *so much* energy that it never comes back at all? We already saw a hint of this in the effective potential picture: it happens exactly when $E \geq 0$. Let's find the critical launch speed for this to happen from some radius R.

The threshold case is $E = 0$ exactly — just barely escaping, arriving at infinity with zero speed left over (like throwing a ball up with just enough speed that it "stops" infinitely far away, never quite falling back, but never having extra speed to spare either). Setting $E = \frac{1}{2}mv_{\text{esc}}^2 - GMm/R = 0$:

$$\boxed{v_{\text{esc}} = \sqrt{\frac{2GM}{R}}.}$$

Compare this to the speed needed for a circular orbit at the same radius, $v_{\text{circ}} = \sqrt{GM/R}$ (which you can get directly from Section 2.4's centripetal-force balance). The ratio is a clean, memorable number:

$$v_{\text{esc}} = \sqrt{2}\, v_{\text{circ}}.$$

This holds at *any* radius, for *any* central mass — a useful fact to keep in your back pocket, since it lets you get escape velocity for free whenever you already know the circular orbital speed, and vice versa.

Putting it all together, the sign of the total energy is really a complete classification scheme for the whole *type* of trajectory a body can have:

$E < 0$	$E = 0$	$E > 0$
ellipse, permanently bound	parabola, marginally escapes	hyperbola, escapes with speed to spare

3 Roche Limit

Here's a question with a slightly unsettling answer: can a moon get *too close* to its planet and simply be torn apart? The answer is yes, and understanding why requires appreciating that gravity isn't uniform — it's slightly stronger on the near side of an object than on the far side, and that difference, small as it usually is, can become catastrophic at close range.

Picture a satellite orbiting close to a planet. The side of the satellite facing the planet is, by definition, a little bit closer to the planet's center than the satellite's own center is; the far side is a little farther. Since gravity falls off with distance, the near side feels a noticeably stronger pull than the center does, and the far side feels a noticeably weaker pull. The result is that the satellite is being *stretched* — pulled apart along the line joining it to the planet, like a piece of taffy being pulled from both ends, even though nothing is grabbing the far side at all; it's simply being left behind relative to how hard the near side is being tugged.

Let's put a number on this stretching. If the satellite has radius r_s and its center is at distance d from the planet (mass M), the near side sits at distance $(d - r_s)$. Expanding to leading order in the (presumably small) ratio r_s/d:

$$g_{\text{near}} = \frac{GM}{(d - r_s)^2} \approx \frac{GM}{d^2}\left(1 + \frac{2r_s}{d}\right).$$

The excess pull, over and above what the satellite's center feels, is the tidal acceleration:

$$\boxed{a_{\text{tidal}} \approx \frac{2GM\,r_s}{d^3}.}$$

Notice this falls off as $1/d^3$, faster than ordinary gravity's $1/d^2$ — which is exactly why tidal effects, while often negligible far away, can become dominant and even destructive at close range. It's a difference-of-forces effect, and differences shrink faster than the forces themselves as you back away.

Now, what's holding the satellite together against this stretching in the first place? Its own self-gravity, pulling surface material back toward its own center:

$$a_{\text{self}} = \frac{Gm_s}{r_s^2}.$$

The satellite survives as long as self-gravity wins. It comes apart exactly when the two are equal — this defines a critical distance, the Roche limit d_R:

$$\frac{2GM\,r_s}{d_R^3} = \frac{Gm_s}{r_s^2} \quad \Longrightarrow \quad d_R^3 = \frac{2M\,r_s^3}{m_s}.$$

This still has the satellite's own size and mass tangled up in it, which isn't the most convenient form. Let's rewrite everything in terms of *mean densities* instead — ρ_M for the planet, ρ_s for the satellite — using $M = \frac{4}{3}\pi R^3 \rho_M$ and $m_s = \frac{4}{3}\pi r_s^3 \rho_s$. Substituting, the factors of r_s^3 cancel completely, leaving a wonderfully clean result:

$$\boxed{d_R = R\left(\frac{2\rho_M}{\rho_s}\right)^{1/3}.}$$

The satellite's actual size dropped out entirely! All that matters is the *contrast* in density between planet and satellite, measured in units of the planet's own radius. A dense, rocky moon can survive much closer to its planet than a loosely-packed, icy or rubble-pile object could — which is exactly the physical reason Saturn's icy rings sit where they do: they're made of material that simply couldn't hold itself together as a single moon any closer in.

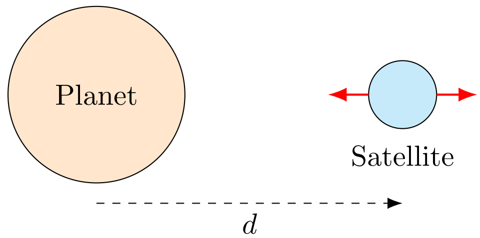

Figure 5: The satellite is stretched along the line joining it to the planet — pulled toward the planet on the near side, and effectively left behind on the far side. Below d_R, this stretching overwhelms the satellite's self-gravity.

4 Barycentre

So far we've been quietly assuming that one body (the "Sun," or the "planet") just sits still while the other orbits around it. That's a fine approximation when one mass hugely outweighs the other, but it's not quite the truth, and it's worth being honest about what's really going on — especially because binary star systems (two comparable-mass stars) show the effect very clearly.

The truth is: *both* bodies orbit. They orbit a shared point called the barycentre, or center of mass, defined the same way you'd define a center of mass for any system:

$$\vec{R}_{\text{cm}} = \frac{M_1\vec{r}_1 + M_2\vec{r}_2}{M_1 + M_2}.$$

If we choose to view the system from a reference frame where this point sits at the origin (a completely legitimate and often very convenient choice), then at every single instant,

$$M_1\vec{r}_1 = -M_2\vec{r}_2.$$

Read this carefully: it says the two position vectors, measured from the barycentre, always point in exactly opposite directions, and their magnitudes are in inverse proportion to the masses. Think of two children of very different weights on a seesaw — for the seesaw to balance, the heavier child has to sit much closer to the pivot than the lighter one. It's exactly the same idea here, except now the "seesaw" is spinning: both stars trace out their own ellipses, similar in shape, but scaled so the heavier star's ellipse is proportionally smaller and hugs closer to the shared center.

This isn't just a geometric curiosity — it's directly useful for astronomers, because we can often measure the individual speeds of two stars in a binary system (via how their light is Doppler-shifted as they move toward and away from us), even when we can't easily measure their individual distances. Since both stars share the exact same orbital period (they have to, since they must remain collinear with the barycentre at all times), and since $r_1/r_2 = M_2/M_1$ always, it follows immediately that

$$\frac{v_1}{v_2} = \frac{r_1}{r_2} = \frac{M_2}{M_1}.$$

Just from a ratio of observed speeds, you get the mass ratio of two stars you may never even be able to resolve as separate points of light. This is one of the quiet triumphs of celestial mechanics:

turning something as simple as "which star's light wobbles more" into a hard number for how much more massive one star is than the other.

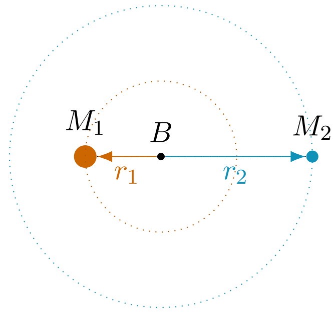

Figure 6: The heavier star (M_1) stays closer to the shared barycentre B; the lighter star (M_2) swings around on a wider circuit. Both always remain on exactly opposite sides of the barycentre.

5 Two-Body Problem

There's something a little unsatisfying about the previous section. We described *where* each star sits relative to the barycentre, but we never actually justified treating the barycentre as if it doesn't move, or explained precisely why we're allowed to reuse all of the single-body machinery from Section 2 (Kepler's laws, vis-viva, and so on) for a system where *both* bodies are moving. Let's fix that properly.

Splitting the Motion in Two

Define the relative separation vector $\vec{r} = \vec{r}_1 - \vec{r}_2$ (how far apart the two bodies are, and in what direction), alongside the barycentre position $\vec{R}_{\text{cm}}$ we already met. Using the relations $\vec{r}_1 = \vec{R}_{\text{cm}} + \frac{M_2}{M_1+M_2}\vec{r}$ and $\vec{r}_2 = \vec{R}_{\text{cm}} - \frac{M_1}{M_1+M_2}\vec{r}$, let's compute the *total* kinetic energy of the system and see what happens:

$$KE = \frac{1}{2}M_1|\dot{\vec{r}}_1|^2 + \frac{1}{2}M_2|\dot{\vec{r}}_2|^2 = \frac{1}{2}(M_1 + M_2)|\dot{\vec{R}}_{\text{cm}}|^2 + \frac{1}{2}\mu|\dot{\vec{r}}|^2,$$

where all the cross-terms between $\dot{\vec{R}}_{\text{cm}}$ and $\dot{\vec{r}}$ cancel out neatly (a small algebraic miracle worth checking for yourself once), and a new quantity has appeared:

$$\boxed{\mu \equiv \frac{M_1 M_2}{M_1 + M_2}}, \qquad \text{the } \textbf{reduced mass}.$$

This is a genuinely beautiful result. It says that the total kinetic energy of *two* moving bodies splits, with no leftover cross-terms, into exactly two independent pieces: the kinetic energy of the total mass $M_1 + M_2$ moving as if concentrated at the barycentre, plus the kinetic energy of a single *fictitious* particle of mass μ, moving with the relative velocity $\dot{\vec{r}}$.

Why This Matters

Because gravity conserves total momentum (Newton's third law guarantees the two gravitational forces on the two bodies are equal and opposite), the barycentre moves in a straight line at constant velocity, forever, completely unaffected by whatever complicated dance the two bodies do around it. All the *interesting* physics — the actual shape of the orbit, how fast the separation changes, the period — lives entirely in the second piece: the relative motion of our fictitious particle of mass μ.

And how does this fictitious particle move? Applying Newton's second law to the relative coordinate directly:

$$\mu\ddot{\vec{r}} = -\frac{GM_1M_2}{r^2}\hat{r} \quad \Longrightarrow \quad \ddot{\vec{r}} = -\frac{G(M_1+M_2)}{r^2}\hat{r}.$$

Notice that μ actually cancels out of the equation of motion entirely (it appears on both sides, since the gravitational force itself is proportional to $M_1M_2 = \mu(M_1+M_2)$) — so this equation is *exactly* the single-body equation of motion we started with all the way back in Section 2, just with the single mass M replaced by the total mass $M_1 + M_2$.

This is the real justification for something we may have been doing on faith so far: every result we derived for a planet orbiting a fixed Sun — Kepler's three laws, the vis-viva equation, the whole effective-potential picture — transfers over immediately and exactly to a two-body system, provided you make one substitution: replace the single central mass M with the total mass $M_1 + M_2$, and understand that a, r, and so on now refer to the *relative* separation between the two bodies, not the distance of one from some fixed point.

An Aside: Why Bother With μ At All?

You might wonder why we introduced this "reduced mass" business at all, since it seems to cancel out of the final equation of motion anyway. The honest answer is: it doesn't cancel out of *everything* — it's essential for correctly computing the system's total energy and angular momentum (both of which genuinely depend on μ, not on $M_1 + M_2$), and it's the rigorous reason we're allowed to treat "two bodies orbiting each other" as equivalent to "one body orbiting a fixed point" in the first place. Physicists reach for this trick constantly, well beyond gravity — it shows up identically in the quantum mechanics of a hydrogen atom, where the electron and proton technically both orbit their shared center of mass too.

6 Lagrange Points

Here's a genuinely delightful question to end on: are there any special places, near a two-body system like the Earth and the Sun, where a third, much smaller object (a spacecraft, an asteroid) could sit and simply *stay put*, neither drifting toward one body nor the other, held in place by the combined tug-of-war of both?

At first, this sounds like it shouldn't be possible except right at the barycentre, in a completely static, non-orbiting situation. But that's not actually the right way to think about it, because our small object doesn't need to be motionless in any absolute sense — it just needs to keep pace with the rotation of the whole system, orbiting the barycentre with exactly the same period as the two main bodies do. If it manages that, then from the point of view of an observer standing on one of the two main bodies (co-rotating with them), the small object would appear perfectly stationary, even though, from far away, everyone involved is actually sweeping around in a circle together.

This is the trick: switch to a **rotating reference frame**, one that spins right along with M_1 and M_2 at their shared orbital angular velocity $\omega = \sqrt{G(M_1+M_2)/a^3}$ (Kepler's Third Law again, doing double duty). In this frame, M_1 and M_2 both look perfectly still, frozen in place on the x-axis. But because we're in a *rotating* frame, we have to add in a fictitious force to account for the rotation — the same centrifugal effect you feel pressed outward against a car door as it turns a sharp corner.

Combining real gravity from both bodies with this fictitious centrifugal effect, we can define an effective potential for the rotating frame:

$$\Phi_{\text{eff}}(x, y) = -\frac{GM_1}{r_1} - \frac{GM_2}{r_2} - \frac{1}{2}\omega^2(x^2 + y^2),$$

where r_1 and r_2 are the distances from our small test object to M_1 and M_2 respectively. A **Lagrange point** is simply a place where this potential is "flat" — $\nabla\Phi_{\text{eff}} = 0$ — meaning a test mass placed exactly there, at rest in this rotating frame, feels no net push in any direction, and so just sits there.

The Three Points on the Line: L_1, L_2, L_3

The easiest points to find are the ones lying directly on the line joining M_1 and M_2 (extended in both directions), since the geometry there is one-dimensional. Let's specialize to the very common and intuitive case $M_2 \ll M_1$ — think of the Sun and the Earth. Somewhere between the two bodies, gravity from the Sun (pulling one way) and gravity from Earth (pulling the other way), together with the centrifugal effect, must all balance. Writing this balance for a small displacement r of this point from Earth, toward the Sun:

$$\frac{GM_1}{(a-r)^2} - \frac{GM_2}{r^2} = \omega^2(a - r).$$

Expanding the left side to leading order in the small quantity r/a, and using $\omega^2 \approx GM_1/a^3$ (an excellent approximation here since $M_2 \ll M_1$), the large GM_1 terms on both sides very nearly cancel each other, leaving behind only the small corrections. What survives, after simplifying, is a beautifully compact result:

$$\boxed{r_{L_1} \approx a\left(\frac{M_2}{3M_1}\right)^{1/3}.}$$

This is the point, between the Earth and the Sun, where the Sun's stronger pull is just barely balanced by the combination of Earth's much weaker (but much closer!) pull and the centrifugal effect. It's precisely why spacecraft like SOHO, which stares continuously at the Sun, are parked at L_1 — close enough to Earth for easy communication, but positioned so they orbit the Sun in lockstep with Earth rather than drifting away.

By an almost identical calculation on the *far* side of Earth (away from the Sun), we find a second point, L_2, at essentially the same distance:

$$\boxed{r_{L_2} \approx a\left(\frac{M_2}{3M_1}\right)^{1/3}.}$$

This is where missions like the James Webb Space Telescope are stationed — shielded from the Sun by Earth, while still keeping pace with Earth's orbit rather than falling behind.

There's a third collinear point, L_3, tucked away on the far side of the Sun, almost exactly opposite Earth. Because it's so far from Earth, Earth's gravity barely perturbs the Sun's field there at all, and the correction to a pure Sun-only orbit is correspondingly tiny:

$$r_{L_3} \approx a\left(1 + \frac{5M_2}{12M_1}\right) \text{ (measured from the Sun).}$$

The Two Off-Axis Points: L_4 and L_5

The remaining two Lagrange points are, in some ways, the most surprising and elegant of the five. They don't sit on the line between M_1 and M_2 at all — instead, picture forming an *equilateral triangle*, using the M_1–M_2 separation a as one side, with the third vertex either leading or trailing M_2 in its orbit. Remarkably, both of these triangular points are equilibria, for *any* mass ratio at all — not just special cases.

Here's the intuition for why. At either triangular point, gravity from M_1 pulls with strength GM_1m/a^2 along one side of the triangle, and gravity from M_2 pulls with strength GM_2m/a^2 along the other side, with exactly a 60° angle between these two directions (fixed purely by the equilateral geometry, regardless of the actual masses involved). If you add these two force vectors together — treating it as a straightforward exercise in vector addition — something rather magical happens: the resultant force points *exactly* toward the barycentre of the M_1-M_2 system, and its magnitude is *exactly* the right amount needed to keep the test mass moving in a circle of that particular radius, at that particular angular velocity ω, around that particular barycentre. Gravity, all on its own, conspires to supply precisely the centripetal force needed — no extra fine-tuning of the mass ratio required. It's a genuine gift from the geometry of the equilateral triangle, combined with how a barycentre divides the line between two masses.

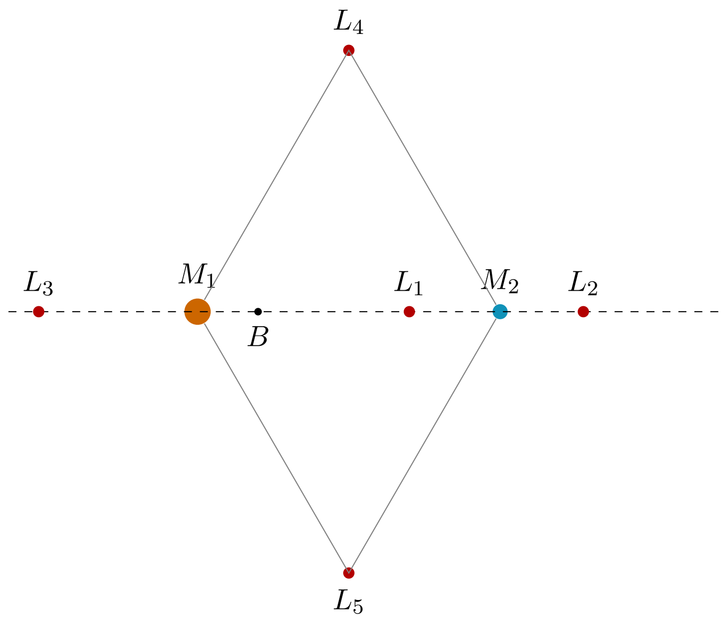

Figure 7: The five Lagrange points. L_1, L_2, L_3 lie strung out along the line joining M_1 and M_2; L_4 and L_5 sit at the apex of equilateral triangles built on the M_1-M_2 separation (drawn here with a comparable-mass ratio for visual clarity — for $M_2 \ll M_1$, the three collinear points crowd close in near M_2 and far out beyond M_1, while L_4/L_5 remain roughly the same distance from the Sun as M_2 itself).

A Note on Stability

It's worth knowing, even without working through the full stability analysis (which involves looking at the curvature of Φ_{eff} in all directions around each point, not just whether the gradient vanishes there), that these five points are not all equally "comfortable" places to sit. L_1, L_2, and L_3 are always **unstable** — nudge an object slightly off one of these points, and it will drift away rather than being pulled back, rather like trying to balance a marble exactly on top of a hill. Spacecraft parked there need occasional small thruster corrections to stay put.

L_4 and L_5, by contrast, *can* be stable — but only if the mass ratio M_1/M_2 exceeds roughly 24.96. Happily, this condition is comfortably satisfied for the Sun-Earth, Sun-Jupiter, and Earth-Moon systems, which is exactly why we observe swarms of Trojan asteroids happily orbiting near Jupiter's L_4 and L_5 points, having sat there stably for billions of years without any need for active correction.

7 Summary Sheet

Topic	**Result**
Newton's law	$F = GMm/r^2$, always attractive
Kepler II (any orbit)	$dA/dt = L/2m =$ const.
Kepler III (circular & elliptical)	$T^2 = 4\pi^2 a^3/GM$
Vis-viva	$v^2 = GM(2/r - 1/a)$
Escape velocity	$v_{\text{esc}} = \sqrt{2GM/R} = \sqrt{2}\, v_{\text{circ}}$
Roche limit	$d_R = R(2\rho_M/\rho_s)^{1/3}$
Barycentre	$M_1 r_1 = M_2 r_2$; $v_1/v_2 = M_2/M_1$
Two-body reduction	$\mu = M_1 M_2/(M_1 + M_2)$; relative orbit uses $M_1 + M_2$
Lagrange L_1, L_2	$r \approx a(M_2/3M_1)^{1/3}$ from M_2
Lagrange L_4, L_5	equilateral triangle with M_1, M_2; stable if $M_1/M_2 \gtrsim 24.96$

8 Further Reading

I mainly wrote this handout to build intuition for celestial mechanics, rather than problem solving capability. For Olympiad-level problem solving and mathematical rigor, use these!

- **Astronomy Olympiad Guide**: A highly optimized learning platform built by former contestants (including IOAA Gold Medalists) to bridge the massive gap between dry university textbooks and competitive problem sheets. It breaks the entire syllabus down into clean, manageable core study tracks.
- **aoXiv Archive**: The definitive comprehensive global repository of astronomy and astrophysics olympiad papers. It hosts an organized, searchable collection of official problems, solutions, and grading schemes from international (IOAA), regional, and national competitions (such as USAAAO, NSEA, and BAAO).
- **Orbital Mechanics Handout** by ongzz: A legendary, mathematically comprehensive deep-dive handout that walks through advanced orbital transformations, coordinate system dynamics, and elite-tier mechanics proofs with spectacular precision.

- **An Introduction to Modern Astrophysics** by Carroll & Ostlie (The "BOB"): The absolute baseline bible for undergraduate astrophysics. Chapters 2 and 10 provide the gold-standard reference for rigorous orbital proofs and binary systems.
- **Problems and Solutions in Astronomy and Astrophysics** by Dr. Aniket Sule: A premium collection of past international competition problems complete with fully unrolled solution keys categorized by subject area.
- **Fundamental Astronomy** by Hannu Karttunen et al.: Excellent for connecting planetary geometry directly to celestial coordinate frameworks and spherical trigonometry expansions.